

// BIOKEY

Your heartbeat is the key — at any stress level.

Stress-Invariant Biometric Authentication · AES-256-GCM

HackTM 2026 // Defence Track

Jigmond Alexandru · Milotin Rareş · Sramec Iosif · Stan Cristian // UPT AC

Authentication fails when it matters most.

The human body changes under stress. Authentication systems don't.

STRESS CHANGES THE BODY

HR: 70 bpm → 160 bpm
ECG enrolled at rest fails in the field.
The system denies its own operator.

CAPTURED HARDWARE

No cryptographic binding to the operator.
Seized device = immediate data access.

COERCION

Passwords and cards can be taken by force.
The operator cannot protect them.

Everything leaves a gap. We close it.

WHAT EXISTS

- × Passwords – coerced or forgotten
- × Smart cards – lost or stolen
- × Single biometric template – fails under stress
- × Hardware encryption – key tied to device, not operator

BIOKEY

- ✓ 5 templates – covers all stress levels
- ✓ Key derived from biometrics – not stored
- ✓ PIN second factor – biometrics alone not enough
- ✓ Captured hardware = only ciphertext

*"We are the key
to your heartbeat."*

AES-256-GCM · HKDF-SHA256 · Multi-Template HRV · PIN Second Factor

github.com/IosifSramec/biokey

HackTM 2026 // Defence Track
